

Ground floor study

Where the garage stair lands, and the floor planned around it: a tall living room behind a pentagon of glass with a corner fireplace at its side and a TV wall over the garage opening, a quarter-turn stair folded into the back corner with the toilet tucked under it, and the metre the house grew at the back given to the living room. Metres, to scale, house frame: x along the house (the valley at the bottom), z across (north to the right).

9.00 × 7.00

over all, a metre further into the hill; 8.40×6.40 clear, 53.8 m^2

5.00 × 5.20

the living room, the whole front from the bedroom wall to the glass: the extra metre is its

x 5.70

where the garage flight lands: the back wall line, 1.30 of landing to the stair up

4 + 3 + 9

the stair up: a quarter turn in the north-west corner, 17×17.1 on 0.27 treads, winders at the corner, the toilet under its west leg

6.40 × 9.10

the panorama: one pentagon of glass, 5.90 at the walls, 9.10 at the apex, the living room's alone

45°

the fireplace: a corner hearth with 1.50 legs on the north wall and the glass, its 2.12 front facing the room diagonally; chimney 0.70 square to 9.90

3.50

the TV wall: the north wall over the garage opening, from the first floor's edge to the hearth; a 75" set at 1.00 to 1.95, 4.4 m from the sofa

26.0 · 9.6 · 5.8 · 2.6

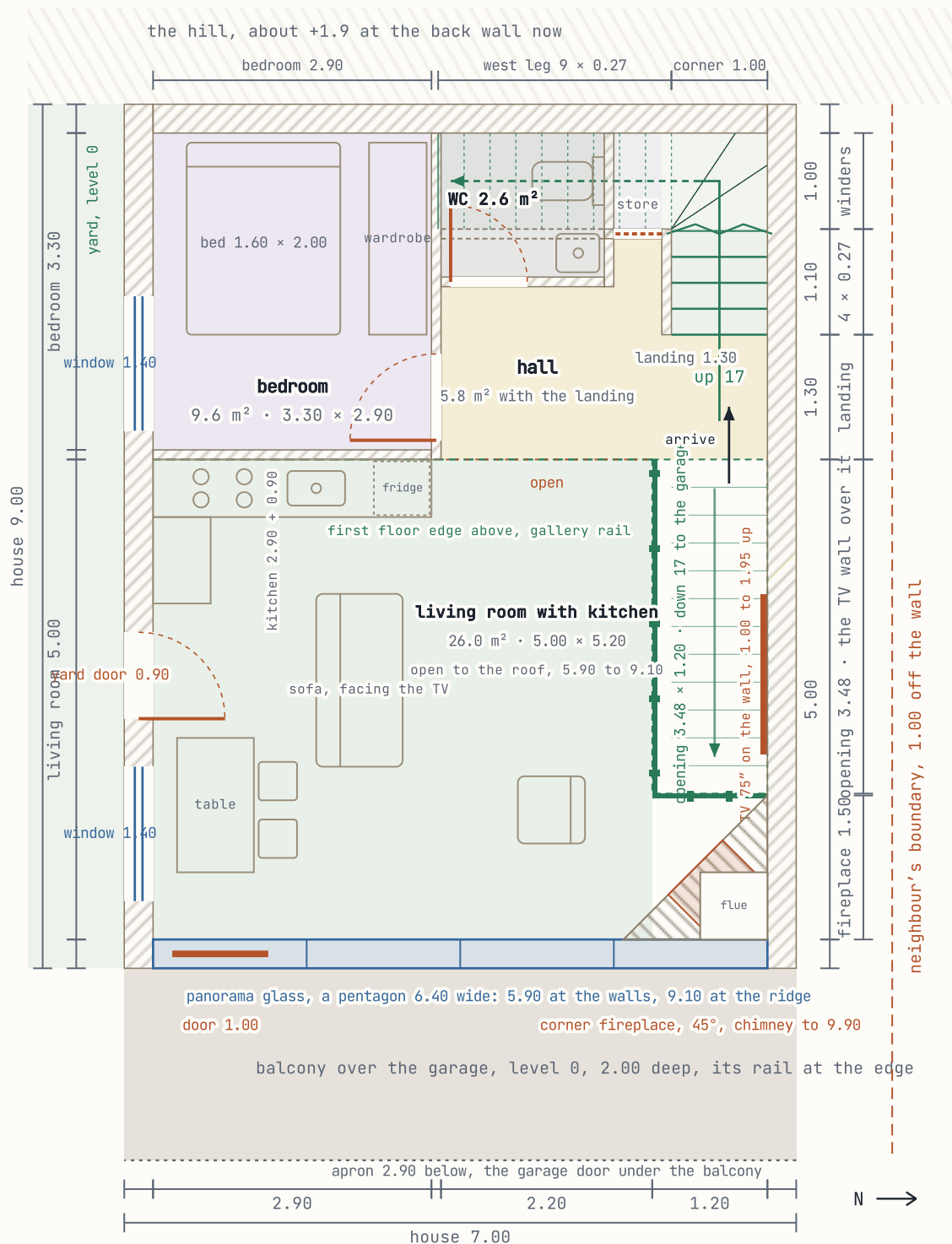
living room with kitchen, bedroom, hall, toilet, m^2

18.3

first floor behind the tall room, m^2 clear: 3.40×6.40 less the stairwell

Plan, 1 m above the floor

The hill is at the top, the valley and the balcony at the bottom, the yard to the left (south), the neighbour's boundary 1 m off the right (north) wall. Walls 0.30, partitions 0.10.



hall
 toilet under the stair
 store under the winders
 bedroom
 living room with kitchen, open to the roof
 glass
 fireplace

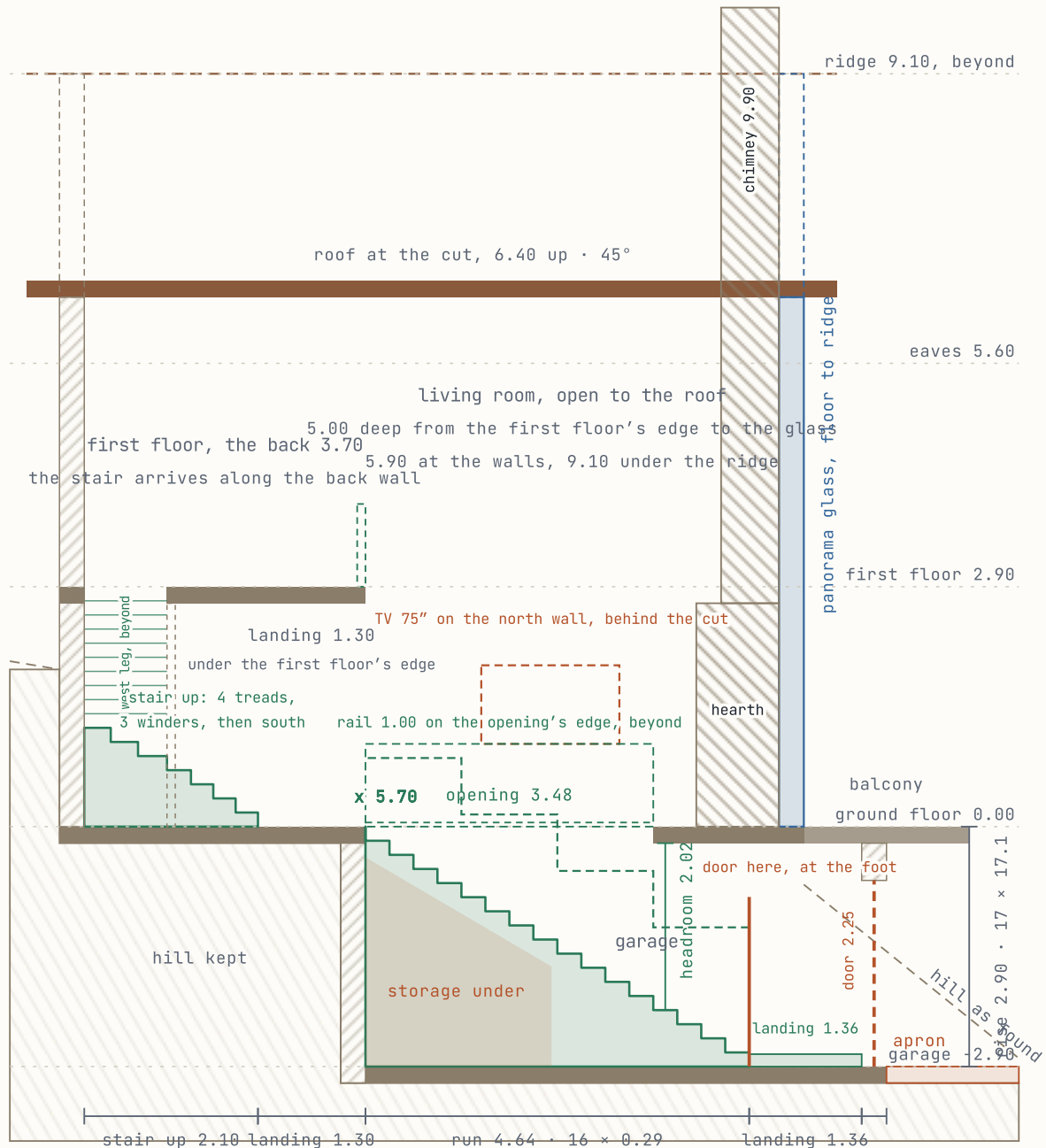
- **The living room has the metre.** It is the whole front from the bedroom wall to the glass, 5.00×5.20 , and it is tall: the first floor stops at $x 5.7$, so from the kitchen to the glass the room rises past the first floor to the roof — 5.90 at the walls, 9.10 under the ridge. Its east wall is the panorama, and nothing

else stands behind it: one pentagon of glass, 6.40 wide, from the floor to the roof, with a 1.00 door to the balcony at its south end.

- **The fireplace** is turned 45° into the north-east corner, between the north wall and the glass: a right-triangle hearth with 1.50 legs, its 2.12 front facing the room diagonally, the 0.80 fire centred on it 0.35 off the floor and 0.50 deep, and a 0.70 × 0.70 chimney in the corner behind it, up through the north slope to 9.90. It covers the glass's first 1.50 at the floor and 0.70 above the ceiling line. The opening's rail turns the corner at x 9.18 to close the end the old hearth closed.
- **The TV wall** is the north wall between the first floor's edge and the hearth, 3.50 m, over the garage opening: a 75" set, 1.67 × 0.95, hangs on it centred on the sofa, 1.40 from the first floor's edge and 0.40 from the hearth, with its bottom at 1.00 — a hair above what the opening's rail hides from the sofa (a seated eye at 1.10, 3.2 m behind the 1.00 rail, sees the wall from 0.96 up); an 85", 1.90 wide, fits as well. Nothing can stand under it, the opening is there, so it is wall-mounted with the cabling in the wall. The sofa faces it 4.4 m back with the fire on its right and the glass beside it; the table moves under the south window.
- **The hall is where the stair lands.** You come up facing the hill at x 5.7, with 1.30 of landing, and the stair up carries straight on: four treads west along the north wall, three winders in the corner, nine treads south along the back wall. South of the landing the hall widens to 1.80 beside the living room, open to it along its whole east side, with the toilet door in its west wall and the bedroom door in its south wall: 5.8 m² with the landing.
- **The stair up** is a quarter turn folded into the north-west corner, 17 risers of 17.1 on 0.27 treads, 1.00 wide, the toilet under its west leg. Its footprint is 4.5 m² of treads, and only 2.2 of that is floor: the flight over the toilet is the toilet's ceiling, 1.61 at its north wall rising to 2.68 at the bedroom wall, and a parapet on the flight's open string, from its underside up to the ceiling, is the toilet's side to the stair. It arrives on the first floor at z 3.73 along the back wall, with a 1.00 landing beyond.
- **The toilet**, 1.50 × 1.70, 2.6 m², is tucked under the west leg with its door from the hall at the full-height end: the pan at the north end under the low part of the flight, 1.6 to 2.0, which is enough to sit under, the basin beside the door at full height, and 2.1 to 2.7 over the floor in front of them. A store under the winders, 1.00 × 0.50 and up to 1.5 high, opens off the hall's bay beside it through a 0.50 hatch in the stair wall. No window: a vent.
- **The entrance from the yard** is a 0.90 glazed door in the south wall beside the kitchen; a 1.40 window further along lights the sofa.
- **The kitchen** stands against the bedroom wall, 2.90 of counter with the fridge at the hall end, and a 0.90 return along the south wall to the yard door. The table sits under the south window, the sofa faces the TV wall and, on its right, the fire.
- **The bedroom** is 3.30 × 2.90 in the south-west corner with a 1.40 window to the yard: a 1.60 × 2.00 bed with its head to the back wall, a wardrobe on its north wall beside the door, 1.20 of floor between the bed and the kitchen wall. Its north wall is the toilet's south wall, so nothing of the stair is over it; its back wall is against the hill.
- **The first floor keeps the back 3.70 m** (x 2–5.7): 3.40 × 6.40 clear less the 3.4 of stairwell, 18.3 m² — a landing, the bathroom over the toilet, a bedroom — with a gallery rail on its edge over the living room, and the attic over it.

Where the garage stair lands: section along it, looking south

Cut 0.8 m from the north wall, through the garage flight. Garage floor -2.90, ground floor 0.00, first floor 2.90, eaves 5.60, ridge 9.10. The roof is cut where it is 6.40 up; the ridge is beyond. The cut passes through the stair up's first leg and winders at the back, and through the hearth and chimney at the glass.

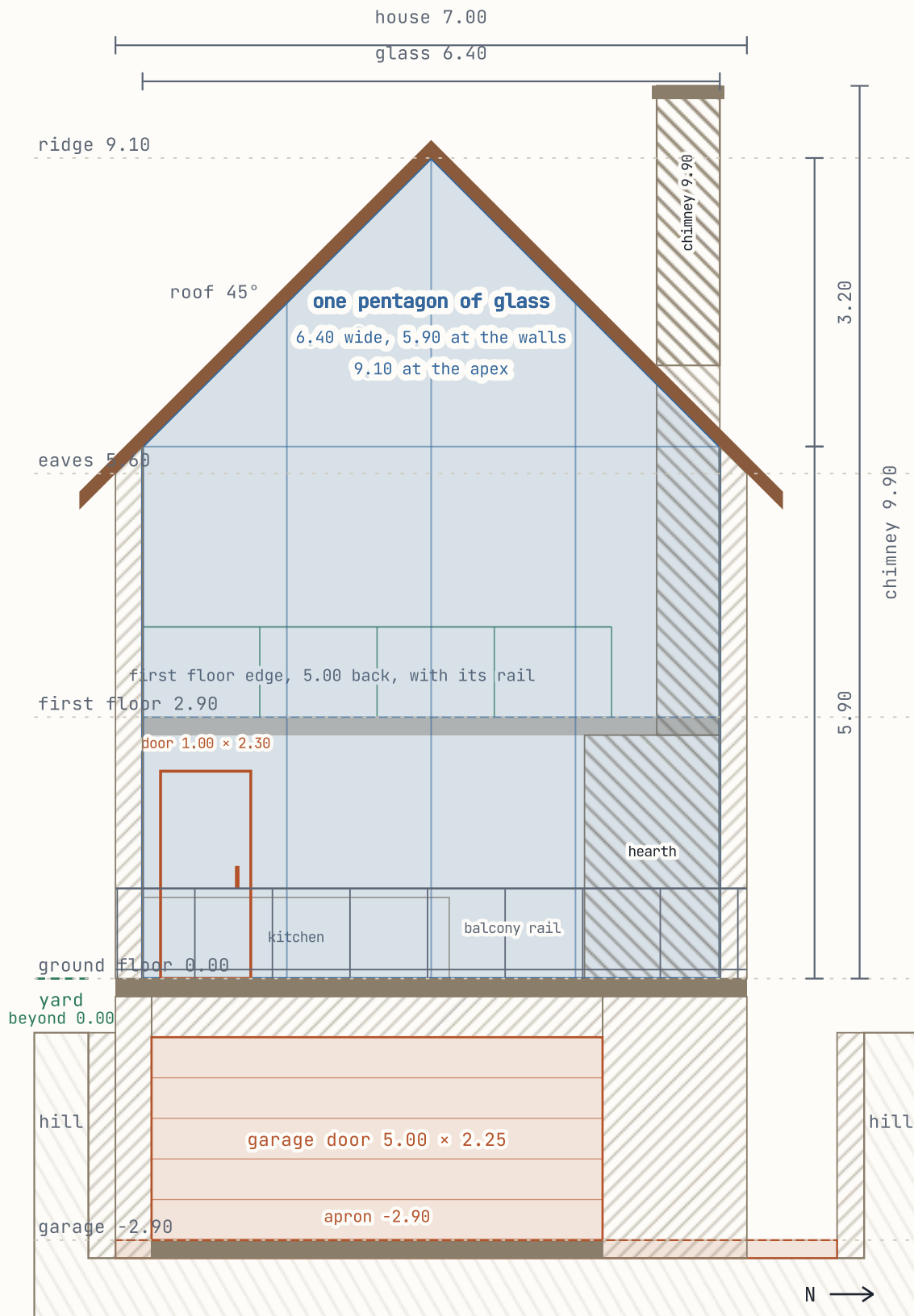


- **The top riser is the garage's back wall line, x 5.7:** the 17th step up lands on the ground floor over the garage's west wall, facing west with 1.30 m of landing before the stair up begins — and the first floor's edge is right above it, so you arrive at the foot of the tall room. The stair up carries the climb on along the north wall, round the corner and south along the back wall.
- **The opening** runs from the 5th riser to the top, 3.48 × 1.20 along the north wall, in the ground floor's deck; above it is the tall room. The deck stays over treads 1 to 4, and the hearth stands on it, closing the opening's east end.
- **Headroom is 2.00** over the 4th tread, the minimum. The stair up has the same: its first four treads sit under the first floor's deck, and from the fifth riser on, at the corner, the deck is open.

- **The toilet is under the west leg**, beyond the cut: the flight is its ceiling, 1.61 at its north wall rising to 2.68 at the bedroom wall, so the pan sits under the low end and the floor in front of it has 2.1 and more; the parapet on the flight's string is its side to the stair.
- **The door between garage and house** belongs at the foot of the garage flight, on the 1.36 landing inside the garage door, swinging onto that landing; a door at the top would open onto the stair.
- **The chimney** rises from the corner hearth through the roof's north slope, where the slope is 6.10 to 6.80 up, and on to 9.90: 0.80 above the ridge, as a flue 2.5 to 3.2 m off the ridge wants. The TV wall is behind the cut, drawn dashed.

East face, from the valley: the pentagon

Looking west at the glass, north to the right as in the plan. The garage door under the balcony, the balcony at the living room's floor, the glass from there to the ridge, the chimney at its right.



- **One pentagon of glass**, 6.40 wide between the north and south walls, 5.90 at the walls, 9.10 at the apex: about 47 m² of glazing, split by mullions every 1.60 and transoms at the first floor's level and the

eaves line.

- **Seen through it:** the hearth and chimney at the north end, the first floor's edge 5.00 back with its rail, the kitchen.
- **The chimney** leaves the north slope near the eaves and stands 9.90 high, 0.80 over the ridge.
- **The balcony door** is at the south end of the glass, 1.00 × 2.30, where the balcony and the yard meet at the house corner.

Rooms

ROOM	CLEAR	M ²	LIGHT AND DOORS
Living room with kitchen	5.00 × 5.20	26.0	open to the roof, 5.90 to 9.10; the glass pentagon east with a 1.00 door to the balcony; the TV wall north, the fireplace turned 45° in the north-east corner; yard door 0.90 and window 1.40 south; open to the hall
Fireplace, 45° in the corner	1.50 legs, 2.12 front	1.8	the hearth 1.1 and the floor in front of it 0.7, between the opening's rail and the fire; chimney 0.70 square to 9.90
Bedroom	3.30 × 2.90	9.6	window 1.40 to the yard; door 0.90 from the hall
Hall: landing and the room south of it	1.30 × 1.10 + 2.30 × 0.50 + 1.80 × 1.80	5.8	the garage flight lands in it, the stair up starts in it; open to the living room
Toilet	1.50 × 1.70	2.6	door 0.80 from the hall; under the stair's west leg, the flight its ceiling from 1.61 to 2.68; no window, mechanical vent
Store under the winders	1.00 × 0.50	0.5	a 0.50 hatch from the hall's bay; 1.2 to 1.5 high
Stair up, on the floor	4 treads + 3 winders	2.2	its west leg is over the toilet and the store, counted there
Garage opening	3.48 × 1.20	4.2	the garage flight below, railed on its south edge
Partitions		1.1	
Inside the outer walls	8.40 × 6.40	53.8	

Still open

- The first floor: 3.40 × 6.40 clear behind the tall room, less the stairwell, 18.3 m² — the stair arrives along the back wall at z 3.73, the bathroom goes beside the

stairwell over the hall, a wall away from the toilet's stack, and the rest is a bedroom; the attic over it is another room if the roof is kept at 45°. A gallery out over the living room would win floor back at the cost of the tall room's depth.

- The fireplace: a wood stove in the same hearth would need a smaller flue; the chimney passes the roof in the corner, 2.5 to 3.2 m off the ridge where the slope is 6.1 to 6.8 up, so it stands up to 3.8 m above the slope and wants bracing.
- The back wall now stands a metre deeper in the hill (about +1.9 of ground against it), and the yard's west edge followed it to x 2: more retaining wall and more cut.
- The roof pitch: 45° drawn, which puts the apex at 9.10. At 35° it is 8.05 and the glass loses a metre.
- 47 m² of east glass: heat in winter and sun in summer want a plan — triple glazing, and shading or a deep roof overhang at the gable.
- The door at the foot of the garage flight: self-closing, fire-rated where the rules ask.
- A north window: the wall is 1 m from the boundary; the rules decide.
- Whether the outer walls sit inside the 9 × 7 (drawn) or outside it: outside would give 9.00 × 7.00 clear and push the east glass onto the balcony's first 0.30.